MILITARY SPECIFICATION SHEET

ELECTRON TUBE TRAVELING-WAVE

TYPE 8364

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Forward-wave amplifier, frequency range 5.4 to 5.9 GHz, 200 mW power output, periodic-permanent-magnet focus, convection cooled, coaxial r-f connectors.

Outline: See figure 1
Mounting position: Any
Cooling: Convection
Connectors, r-f: See figure 1
Connectors, power: See figure 1
Focusing circuit: PPM integral
Weight: 3 pounds

ABSOLUTE RATINGS: See notes 1 and 2

Parameter:		Ef	If	Ec	Ic	Ew	Ik
Unit:		V	A surge	V	mA	V	mA
Maximum:		11.5	4.0	30	1.0	1350	15
Minimum:		10.5	---	-250	---	---	---
TEST CONDITIONS:	(1):	11.0	---	Note 5	---	Note 5	---
	(2):	0	---	0	---	0	---

Parameter:		Pp	Pi(rf)	tk	TA	TE
Unit:		W	dbm	sec	°C	°C
						Note 3
Maximum:		17	20	---	75	150
Minimum:		---	---	60	-55	-55
TEST CONDITIONS:	(1):	---	Note 25	---	25±5	---
	(2):	---	0	---	---	---

GENERAL:

Qualification - Required
Marking - See note 4

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Method	Requirement or Test	Conditions	Symbol	Limits		Unit
				Min	Max	
	Qualification inspection (Note 26)					
---	Temperature	Test condition (1); F=5.9 GHz Note 10	---	---	---	---
---	Low-temperature turn-on	Test condition (1); F=5.9 GHz Note 11	---	---	---	---
---	Humidity	Test condition (1); F=5.9 GHz Note 12	---	---	---	---
---	Temperature, non-operating	Test condition (2) Note 13	---	---	---	---
---	Initial life test	Test condition (1) F=5.9 GHz, Notes 19, 23, 24, 28				
	Quality conformance inspection, part 1 (See Note 27)					
1301	Heater current	EF=11.0 V	If	0.73	1.05	A
4250	Large-signal Power output	Test condition (1):	Pi(rf)	-16	---	dbm
		F=5.4, 5.5, 5.6, 5.7, 5.8, 5.9 GHz Notes 6, 7, 8, 21	Pso	23	---	dbm
		F=5.4 to 5.9 GHz	ΔPo	---	2.0	dB
		F=5.4 to 5.65 GHz	ΔPo	---	1.5	dB
		F=5.65 to 5.9 GHz Note 9	ΔPo	---	1.5	dB
			Ew	1100	1250	V
			Ec	-9	0	V
4260	Noise figure	Test condition (1): F=5.4, 5.65, 5.9 GHz	NF	---	35	dB
1256	Control-electrode current	Test condition (1): Pi(rf)=0	Ic	0	0.5	mA
1256	Cathode current	Test condition (1): Pi(rf)=0; tk=60 sec	Ik	6	10	mA
	Quality conformance inspection, part 2 Notes 14, 15					
4261	Spurious Power Output (1)	Test condition (1); Pi(rf)=0; F=4.9 to 6.4 GHz Note 16	spo	---	-10	dbm

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Method	Requirement or Test	Conditions	Symbol	Limits		Unit
				Min	Max	
	Quality conformance inspection, part 2 Notes 14, 15 (continued)					
4261	Spurious Power output (2)	Test condition (1): Pi(rf)=0; F=4.9 to 6.4 GHz Note 16	spo	---	-10	dbm
4250	Power output variation	Test condition (1): F=5.4, 5.9 GHz; Ef=10.5 to 11.5 V Note 17	ΔPo	---	1.0	dB
		ΔEw=±1% Note 18	ΔPo	---	1.5	dB
4266	Direct interelectrode Capacitance	Note 10	Cc-all	---	50	pf
	Vibration	Test condition (1): F=5.65 GHz Note 21	---	---	---	---
1042	Shock	Test condition (2): Note 22	---	---	---	---
	Quality conformance inspection, part 3 Note 20					
---	Life test	Test condition (1): Group S F=5.9 GHz Notes 23, 24	t	5000	---	hrs
---	Life test end point	Test condition (1): F=5.9 GHz Pi(rf)=0 dbm, max	Po	20	---	dbm

NOTES:

1. Typical operating conditions for this device are described by Test conditions (1). Because of the number of interrelated conditions possible with traveling-wave tubes, Ratings shall refer to each individual rating with the tube otherwise operating in the above test condition. Note that Ratings are not of themselves test conditions or operating conditions.

2. Cathode and one leg of heater shall be connected internally. Shell, helix, and collector shall be at ground potential. All voltages are referred to cathode.

3. TE is the temperature measured at point "A" on Figure 1.

4. Each tube shall be marked with the following information:

   (a) Serial number
   (b) Cathode voltage for large-signal operation, Ew
   (c) Grid voltage for large-signal operation, Ec
   (d) Cathode current for large-signal operation
   (e) Input and output r-f connector identification

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5. Ec and Ew under Test condition 1 are defined as the voltages at which the tube produces large-signal power output within the specified limits. These voltages shall be identical with the values specified in Note 4.

6. A filter shall be inserted between the tube and the power-measuring device to attenuate harmonic-frequency components by not less than 30 dB relative to the fundamental frequency.

7. At each of the frequencies specified, with Pi(rf) adjusted to obtain saturated power output, Pso, the values of Pi(rf) and Pso shall be within the limits specified. Vary Pi(rf) above and below the saturation value to obtain the Pso-3dB points. The values of Pi(rf) at these points shall be within the specified limits and shall differ by at least 9 dB.

8. Adjust Pi(rf) to obtain saturated power output. Using this value of Pi(rf), measure Po continuously across the band 5.4 to 5.9 GHz. The total variation in power output, ΔPo, across the band shall be not greater than the limit specified.

9. Cc-all is the capacitance between grid and all other elements, including the shell. The grid lead shall be at maximum separation from the shell and all other leads.

10. The temperature test shall be conducted in accordance with the requirements of MIL-E-16400(NAVY), for class 4 equipment. The tube shall operate within the limits specified under Quality conformance inspection, Part 1, Large-signal power output.

11. With the tube in the environment chamber, reduce the chamber temperature to 0±2 degrees Centigrade, and maintain the chamber temperature for at least 24 hours with the tube in the non-operating condition. At the end of the 24-hour period, turn on the tube. The tube shall operate within the limits specified under Quality conformance inspection, Part 1, Large-signal power output, within 25 minutes of application of heater voltage.

12. The humidity test shall be conducted in accordance with the requirements of MIL-E-16400(NAVY), for class 4 equipment. Reference measurements and all subsequent measurements shall consist of the test for Large-signal power output under Quality conformance inspection, Part 1.

13. The non-operating temperature test shall be conducted in accordance with the requirements of MIL-E-16400(NAVY). Upon completion of the test the tube shall operate within the limits specified under Quality conformance inspection, Part 1, Large-signal power output.

14. The Quality conformance tests, Part 2, shall be performed in accordance with MIL-STD-105, Inspection level 2, with an AQL of 4%. In the event of lot rejection, tightened inspection shall be invoked. Normal inspection shall be resumed when two consecutive lots have conformed with the tests of Quality conformance inspection, Part 2. For purposes of inspection, the lot size shall be one month's production.

15. Tubes shall be selected a random from the production lot, and shall previously have passed the tests of Quality conformance inspection, Part 1.

16. For Spurious power output (1) the output termination shall have a nominal value of 50 ohms, and the input shall be terminated in a variable-phase short circuit. For spurious power output (2) the terminations shall be transposed. A preselector having a passband of not more than 30 MHz, and a tuning range of 4.9 to 6.4 GHz shall be used between the tube and the power-measuring device. The spurious power output, spo, shall be not greater than the specified value in any portion of the above frequency range at all phases of the short circuit.

17. At each of the specified frequencies adjust the r-f input power Pi(rf) to obtain saturation. Reduce the heater voltage to the lower of the specified values and read the power output, Po. Increase the heater voltage to the higher of the specified values and read the power output. Allow the power to stabilize at each reading. The total change in output shall be within the specified limit.

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18. At each of the specified frequencies adjust Pi(rf) to obtain saturation. Reduce Pi(rf) by 4dB. Vary Ew from Ew-1% to Ew+1% and note the power output, Po, at each value. Increase Pi(rf) to the saturation value plus 4dB and repeat the power measurements. The change in power, ΔPo, from the reference value, Po, at Ew, in each case shall be within the limit specified.

19. Tubes for these tests shall be selected at random from the production lot, and shall previously have passed the tests of Quality conformance inspection, Part 1.

20. The tube shall meet the power output conditions at all frequencies within the band when operating into a load of 1.5:1 VSWR, minimum.

21. The tube shall be rigidly mounted to the vibration table and shall be vibrated in accordance with the requirements of MIL-STD-167 (SHIPS), Type 1, Environmental vibration. The detected output of the tube shall be displayed on an oscilloscope which shall be calibrated prior to vibration in the d-c mode using slow sweepby means of a calibrated attenuator. The face of the display tube shall be marked with lines corresponding to the saturated power output level and to levels ±0.5dB therefrom. The attenuator then shall be returned to the saturated level. The displayed variations attributable to vibration shall be within the calibration lines.

22. The shock test shall be conducted in accordance with MIL-STD-202, method 213, Shock (Specified Pulse), using the sawtooth shock pulse, Figure 213-2, with peak acceleration of 40g, and duration of 10 msec. The tube thereafter shall be tested for Quality conformance inspection, Part 1, Large-signal power output, and shall operate within the specified limits.

23. Tubes selected for this test shall previously have passed the shock and vibration tests.

24. The tube shall be operated intermittently according to the following schedule for the duration of the life test:

Heat voltage, Ef, only	4 hours
Test condition (1)	8 hours

Life hours will be based on filament hours.

25. Pi(rf) is the incident power to the input r-f terminal of the tube. Under test condition 1 unless otherwise specified, Pi(rf) shall be that rf input power required to obtain saturation power output, Pso, at the specified frequency.

26. Three tubes are required for these tests. Quality conformance inspection, Part 1, shall be performed prior to and upon completion of Qualification inspection.

27. All tubes shall be tested in accordance with Quality conformance inspection, part 1.

28. Initial Groups life test shall be conducted for 5,000 hours in the following sequence:

(a) Operate in accordance with Quality conformance inspection, part 3 for 336±36 hours.
(b) Test in accordance with Quality conformance inspection, part 1 and Quality conformance inspection, part 2, Power output vibration.
(c) Operate in accordance with Quality conformance inspection, part 3 for 72 hours at F=5.9GHz and simultaneously operate in accordance with Quality conformance inspection, part 2.

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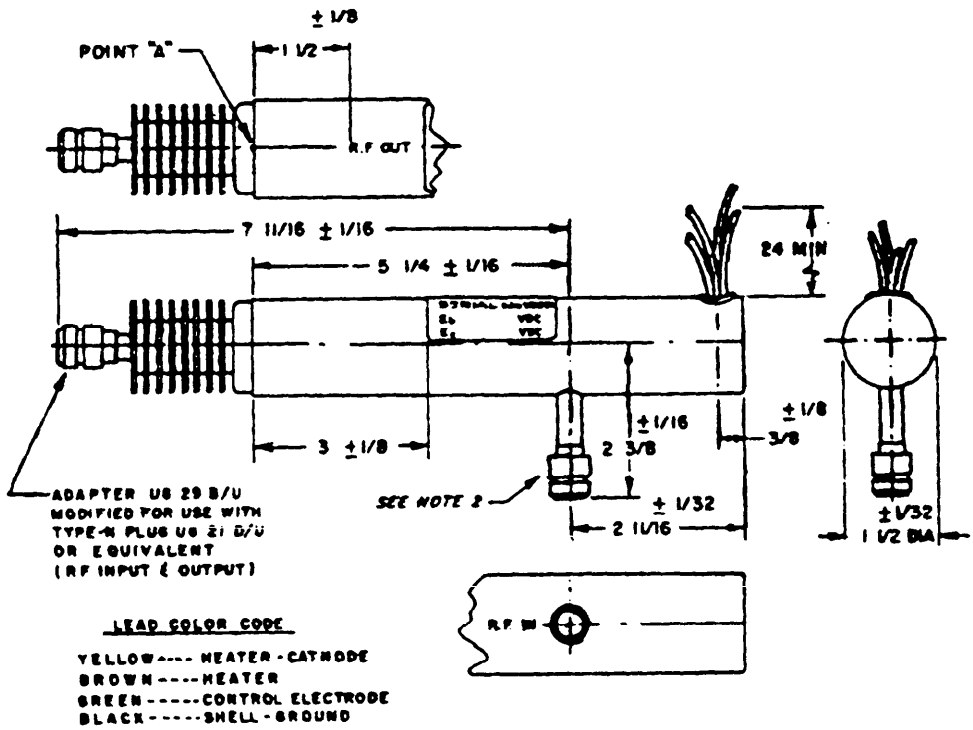
The "Endurance test" of MIL-STD-167 (SHIPS) shall apply. The vibration frequency shall be the resonant value determined from Quality conformance inspection, part 2, Vibration test, or 33 Hz, with amplitude as shown in MIL-STD-167, Table 1.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

(Project 5960-N901)

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NOTE: 1. Point "A" = Location where envelope temperature is measured
2. Maximum permissible dia. of any portion of RF input = .740

FIGURE 1. Outline drawing of electron tube, type 8364.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least co- Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, ar ·· to preparing activity. Comments and suggestions submitted on this form do not constitute or imply Author· ·tion to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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